

Orbis

An Intelligent Academic Support System with AI and RAG

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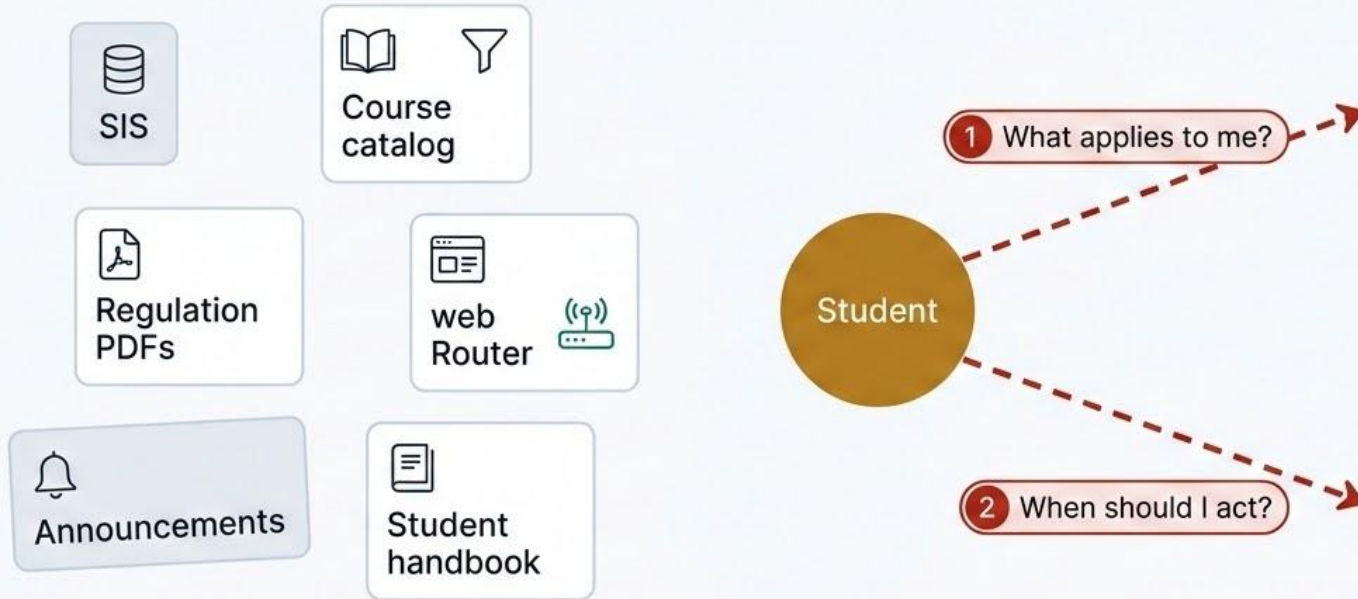
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Turning scattered information into actionable, traceable guidance.

Problem

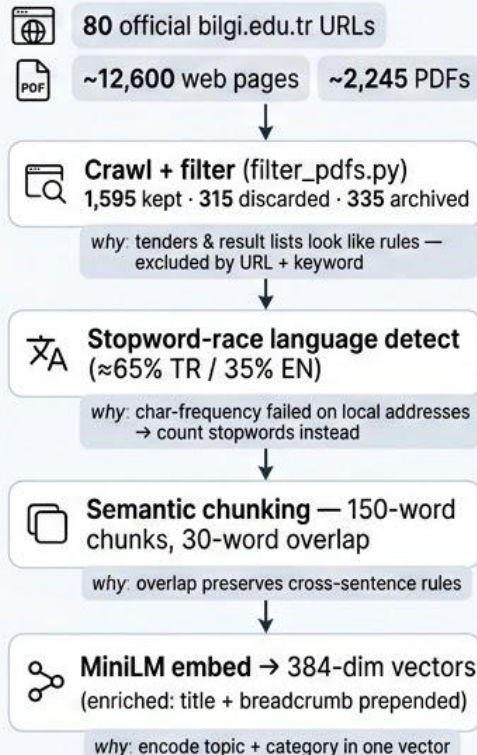


- Rules are scattered across many systems
- Students don't always know what to ask
- Portals store records, they don't interpret them
- The hardest obligations are the ones you never think to ask about

Orbis: Detailed System Showcase

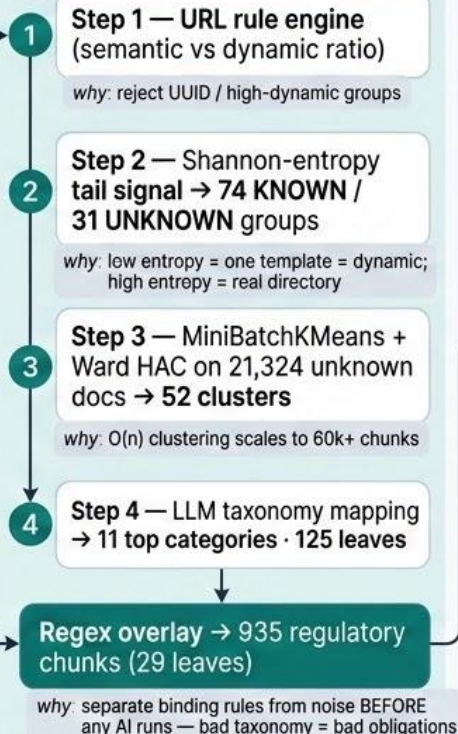
BAND A

1 · Data collection & ingestion



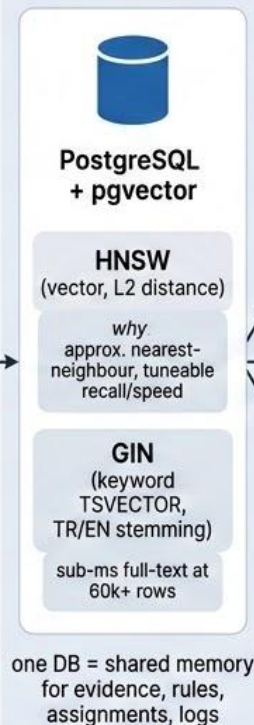
BAND B

2 · Taxonomy guardrail (4 steps)



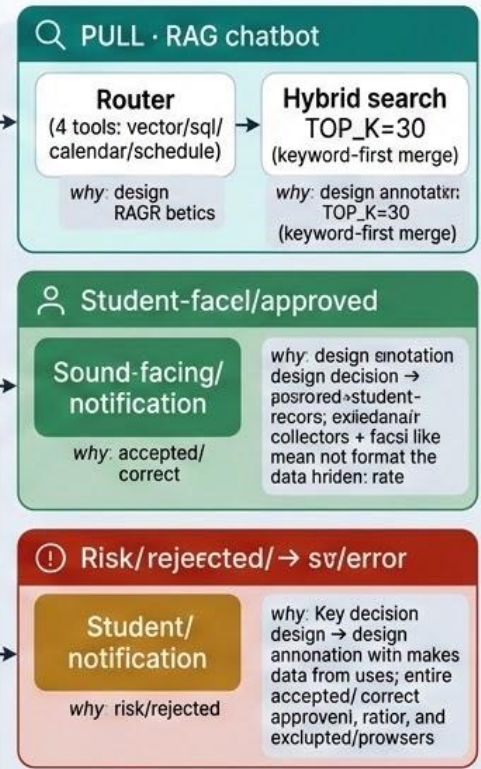
BAND C

3 · Shared store



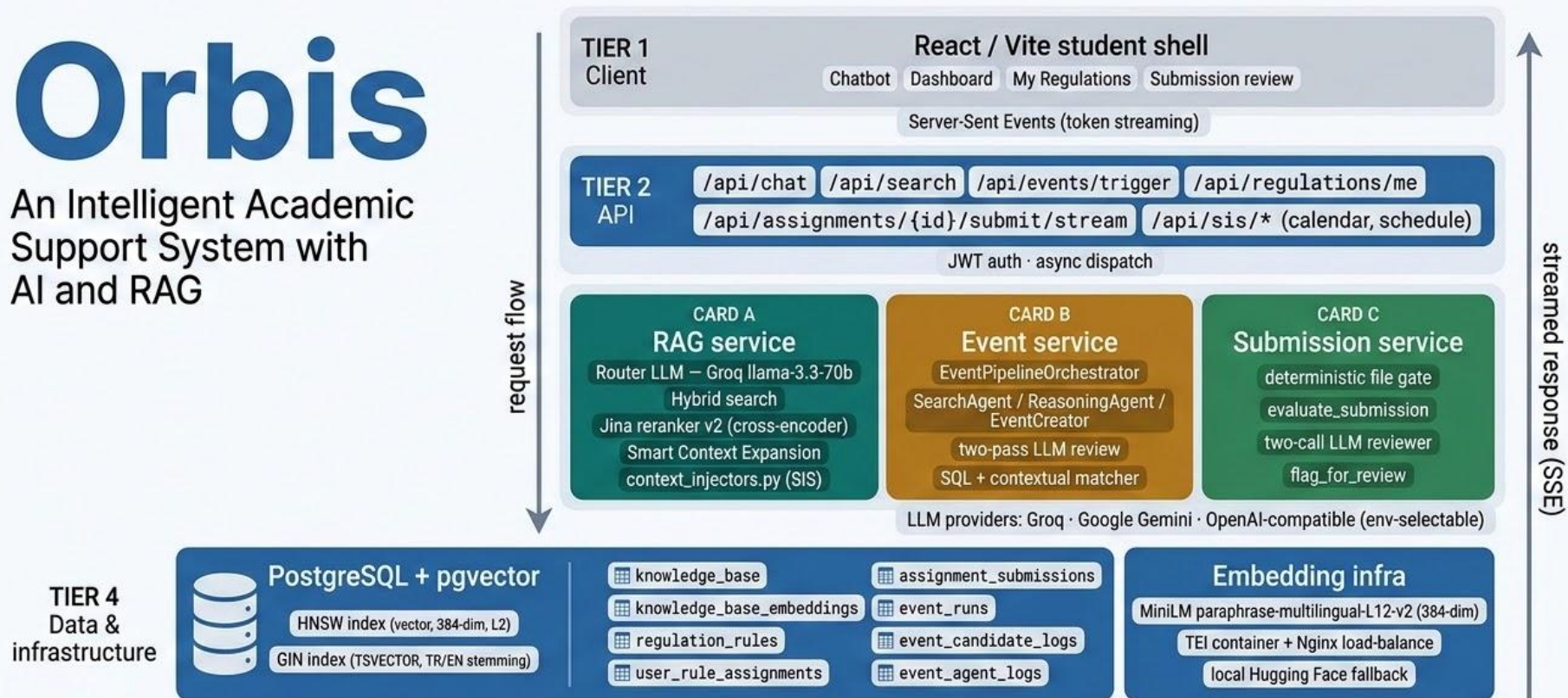
BAND D

4 · Three runtime paths → student



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One PostgreSQL + pgvector instance is the shared memory for all three paths — relational schema and vector index in the same database

Implementation and Challenges

TOP ZONE: Progress in Senior Design Project II

built on the S1 foundation:
scraping, embedding choice, DB



Backend + all API
routes completed



Taxonomy
(125 leaves)



RAG chatbot
end-to-end



Event pipeline
+ matching



Submission
review agent



Gold-label
evaluation harness

BOTTOM ZONE: Challenge

How we solved it

Evidence



Formal rules mixed with
casual news/announcements



4-step taxonomy guardrail isolates
rules before any AI runs

Evidence

935 regulatory chunks / **60,649**



Vague, non-actionable
obligations



Two-pass adversarial LLM review
(pass 2 removes stale/weak rules)

Evidence

55% accepted · **45%** rejected
by design



65% of corpus is temporal
news/events drowning answers



Keyword-first hybrid search + 2-stage
Jina rerank + Smart Context Expansion

Evidence

0 temporal contamination
in final context



Multi-article regulations
fragmented across 150-word chunks



Smart Context Expansion rebuilds
full source document around top hits

Evidence

8-step grade-appeal procedure
recovered from 4 chunks



Prompt injection hidden
inside uploaded files



Two-call reviewer treats document
as evidence, not instruction

Evidence

injection resistance **1/4** → **4/4**



Every decision is logged and auditable (event_candidate_logs · event_agent_logs) (~18pt)

Evaluation dashboard: RAG, event extraction, assignment matching

RAG chatbot — 100-query bilingual benchmark, 7 routing categories

92.3%
Routing accuracy

94.7%
Context recall

0.94
MRR

97.6%
Semantic accuracy

Embedding benchmark
70.8% **0.571**
Best top-3 Best MRR
(TEI MiniLM, 100-word)
justifies model + chunk size

Event extraction

323 candidates · gold-labeled

0.978
Precision

0.574
Recall

0.723
F1

0.588
Accuracy

Confusion (N=323)

Accept	174 TP	4 FP	Very precise — avoids bad obligations
Reject	129 FN	16 TN	

Assignment matching

712 pairs = 4 profiles × 178 obligations

0.608
Precision

0.446
Recall

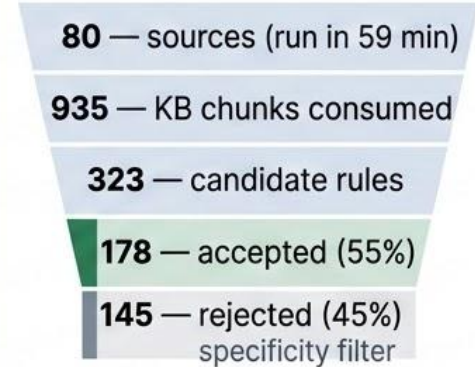
0.514
F1

Per profile (P / R / F1)

P1	0.548 / 0.469 / 0.505
P2	0.667 / 0.358 / 0.466
P3	0.650 / 0.500 / 0.565
P4	0.553 / 0.457 / 0.500

Weakest
layer — clear
next target

Pipeline funnel



Submission review

32 cases = 4 assignments × 8 strata

		Confusion (N=32, positive=approve)	
93.8% Accuracy	1.000 Recall	Approve gold	
		8 TP	0 FN
0.889 F1	0.800 Precision	Reject gold	
		2 FP	22 TN

Zero false negatives — no genuine work rejected

Ablation — single-call → two-call agentic

metric	before	→	after
Accuracy	0.781	→	0.938 ↑
Injection resistance	1/4	→	4/4 ↑

two-call agent verifies each requirement, bypassing embedded instructions

How we tested — 8 strata (4 each)

☐ approve — 4/4	☐ reject empty — 4/4
☐ approve multi-file — 4/4	☐ reject corrupt — 4/4
☐ reject wrong content — 4/4	☐ reject injection — 4/4
☐ reject wrong extension — 4/4	☐ flag partial — 2/4

7 of 8 strata perfect

Prompt-injection resistance

4 / 4

tested injection cases rejected

Directional finding — not a general security guarantee

From static lookup to **proactive**, **inspectable**, **measurable** support

- ✓ **Three paths shipped:** grounded search, proactive obligations, submission review
- ✓ **Boundaries kept:** advisory not authoritative; screening not grading
- ✓ Every subsystem is **measurable** — results point straight at the next work

Future-work slide



Improve matching recall

Recall-oriented second pass on the weakest layer



Live SIS integration

Replace synthetic profiles with real academic state



Human review queue

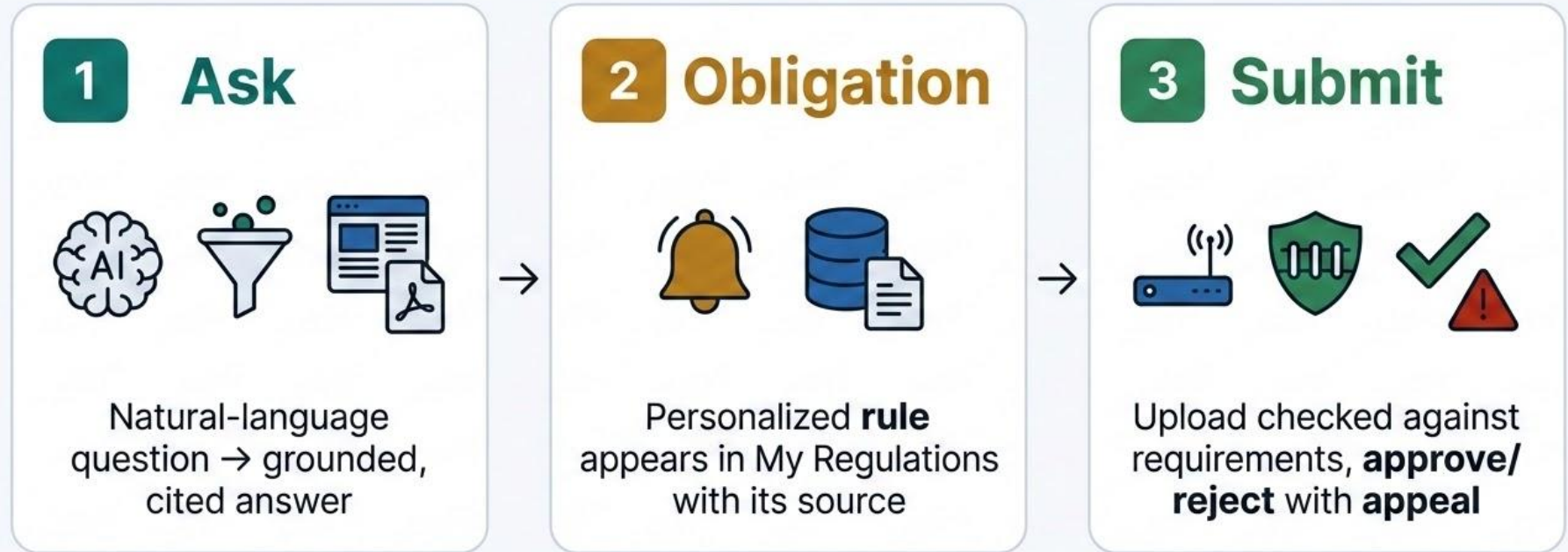
Explicit flag-for-review class for partial/suspect work



Broader safety testing

Injection patterns beyond the tested cases

Demo storyboard



≈30pt Proactive, inspectable, measurable academic support

